

Piecewise potential-vorticity inversion on the sphere

Method, step by step

`pv_inversion_spherical`

Abstract

This document states the equations the package solves, the steps by which it solves them, and how the calculation differs from the limited-area Fortran chain that solves the same system. Every number quoted comes from the example runs that ship with the package.

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1 What is solved

Vertical coordinate. The Exner function,

$$\Pi = C_p \left(\frac{p}{p_0} \right)^\kappa, \quad \kappa = \frac{R_d}{C_p} = \frac{2}{7}, \quad (1)$$

with $C_p = 1004.5 \text{ J kg}^{-1} \text{ K}^{-1}$, $R_d = 287.0 \text{ J kg}^{-1} \text{ K}^{-1}$, $p_0 = 1000 \text{ hPa}$. Π decreases upward and equals C_p at 1000 hPa. In this coordinate hydrostatic balance is linear in the potential temperature, and its Π -derivative is the static stability:

$$\frac{\partial \Phi}{\partial \Pi} = -\theta, \quad S = -\frac{\partial \theta}{\partial \Pi} = \frac{\partial^2 \Phi}{\partial \Pi^2}. \quad (2)$$

Horizontal operators. With λ, φ longitude and latitude and $a = 6.371 \times 10^6$ m, $\partial_x = (a \cos \varphi)^{-1} \partial_\lambda$, $\partial_y = a^{-1} \partial_\varphi$, and for a vector (F_x, F_y) and a scalar s

$$\begin{aligned}\nabla \cdot \mathbf{F} &= \partial_x F_x + \partial_y F_y - \frac{\tan \varphi}{a} F_y, & \mathbf{k} \cdot \nabla \times \mathbf{F} &= \partial_x F_y - \partial_y F_x + \frac{\tan \varphi}{a} F_x, \\ \nabla^2 s &= \partial_x \partial_x s + \partial_y \partial_y s - \frac{\tan \varphi}{a} \partial_y s.\end{aligned}\quad (3)$$

The wind is the rotational wind of a streamfunction, $\mathbf{V} = \mathbf{k} \times \nabla \psi$, so $u = -\partial_y \psi$, $v = \partial_x \psi$ and $\zeta = \nabla^2 \psi$. The Coriolis parameter that enters the operator is the floored, equator-symmetric f^* of Step 1.

Potential vorticity. Ertel's $q = \rho^{-1} \boldsymbol{\eta} \cdot \nabla \theta$, made hydrostatic and written on Exner surfaces, is (2.2) of Davis and Emanuel (1991),

$$q = -\frac{g\kappa\Pi}{p} \left(\eta \frac{\partial \theta}{\partial \Pi} - \frac{1}{a \cos \varphi} \frac{\partial v}{\partial \Pi} \frac{\partial \theta}{\partial \lambda} + \frac{1}{a} \frac{\partial u}{\partial \Pi} \frac{\partial \theta}{\partial \varphi} \right), \quad (4)$$

$\eta = f + \zeta$ the vertical absolute vorticity. Replacing the shear of the total wind by that of the rotational wind and eliminating θ with (2) gives their (2.3). Divided through by the leading factor, the field the inversion works with is

$$\hat{q} = q \frac{p}{g\kappa\Pi} = (f^* + \nabla^2 \psi) \frac{\partial^2 \Phi}{\partial \Pi^2} - \nabla \left(\frac{\partial \psi}{\partial \Pi} \right) \cdot \nabla \left(\frac{\partial \Phi}{\partial \Pi} \right), \quad \frac{p}{g\kappa\Pi} = \left(\frac{p}{p_0} \right)^{1-\kappa} \frac{p_0}{\kappa g C_p}. \quad (5)$$

This factor is the whole of the unit conversion; everything else in the solver is SI. Values are quoted in PVU, $1 \text{ PVU} = 10^{-6} \text{ m}^2 \text{ K kg}^{-1} \text{ s}^{-1}$.

Balance. Charney (1955, p. 24) writes the divergence equation, with the divergent wind omitted, as

$$\nabla^2 \psi + \nabla \psi \cdot \nabla f - \frac{2}{f} \left[\left(\frac{\partial^2 \psi}{\partial x \partial y} \right)^2 - \frac{\partial^2 \psi}{\partial x^2} \frac{\partial^2 \psi}{\partial y^2} \right] = \frac{\nabla^2 \phi}{f}, \quad (6)$$

which multiplied through by f is (1.3) of Davis (1992), and in spherical coordinates (2.1) of Davis and Emanuel (1991). It is close to gradient-wind balance and is solvable for ψ where $\nabla^2 \phi / f + f/2 > 0$. The package solves it in the exact spherical form obtained by taking the divergence of the momentum equation on the sphere,

$$\nabla^2 \Phi = N(\psi), \quad N(\psi) = \nabla \cdot [(f^* + \zeta) \nabla \psi] - \frac{1}{2} \nabla^2 |\nabla \psi|^2, \quad (7)$$

in which every term is a divergence or a Laplacian of a product formed on the grid. On the sphere this differs from the plane form by the curvature term,

$$\nabla \cdot (\zeta \nabla \psi) - \frac{1}{2} \nabla^2 |\nabla \psi|^2 = 2 \det(\text{Hess } \psi) - \frac{|\nabla \psi|^2}{a^2}. \quad (8)$$

The quadratic part of N comes from the symmetric bilinear form

$$B(a, b) = \nabla \cdot (\zeta_a \nabla b) + \nabla \cdot (\zeta_b \nabla a) - \nabla^2 (\nabla a \cdot \nabla b) = \zeta_a \zeta_b - (D_{1a} D_{1b} + D_{2a} D_{2b}) - \frac{2 \nabla a \cdot \nabla b}{a^2}, \quad (9)$$

D_1, D_2 the two deformation components, so that the tangent of the balance row at a reference $\tilde{\psi}$ is $\nabla \cdot (f^* \nabla \psi') + B(\tilde{\psi}, \psi')$ and its principal symbol has eigenvalues $f^* + \tilde{\zeta} \mp \tilde{D}$, $\tilde{D} = (\tilde{D}_1^2 + \tilde{D}_2^2)^{1/2}$. The row is elliptic where the absolute vorticity exceeds the deformation, $(f^* + \tilde{\zeta})^2 > \tilde{D}^2$, which is Charney's condition in another form. $\tilde{D}$ is read off (9) with $a = b$, from spectral operations alone.

The coupled system. On each interior level the two relations hold together, for the two fields (Φ, ψ) :

$$\nabla^2 \Phi = \nabla \cdot [(f^* + \nabla^2 \psi) \nabla \psi] - \frac{1}{2} \nabla^2 |\nabla \psi|^2, \quad (10)$$

$$\hat{q} = (f^* + \nabla^2 \psi) \frac{\partial^2 \Phi}{\partial \Pi^2} - \nabla \left(\frac{\partial \psi}{\partial \Pi} \right) \cdot \nabla \left(\frac{\partial \Phi}{\partial \Pi} \right). \quad (11)$$

The balance row (10) couples Φ and ψ on each level; the potential-vorticity row (11) couples the levels vertically through Φ_{III} and the shear terms. Davis and Emanuel (1991, p. 1934) state that (2.1) and (2.3) form a complete system for Φ and Ψ given q . Both rows are quadratic: (10) in ψ , (11) bilinear in (ψ, Φ) . The system is elliptic where $f^* + \zeta > 0$, $S > 0$ and the balance row's condition above holds.

Boundary conditions. At the top and the bottom,

$$\frac{\partial \Phi}{\partial \Pi} = -\theta_B \quad (\text{lower half level}), \quad \frac{\partial \Phi}{\partial \Pi} = -\theta_T \quad (\text{upper half level}), \quad \frac{\partial \psi}{\partial \Pi} = -\frac{\theta - \langle \theta \rangle}{f^*} \quad (\text{both}), \quad (12)$$

$\langle \cdot \rangle$ the area mean. The streamfunction condition is the thermal-wind form of the hydrostatic one, $\Phi \simeq f\psi$, with the horizontally uniform part of θ removed because it carries no flow, and the division by f^* made before any horizontal gradient is taken. There is no lateral boundary: the domain is the whole sphere, and the only additional conditions are the gauges of Step 3.

The levels. Nine pressure levels. The two outermost carry boundary potential temperature on the half levels next to them (near 925 and 140 hPa); the seven between carry potential vorticity and are the unknowns.

p [hPa]	Π [J kg ⁻¹ K ⁻¹]	carries	δ_k [J kg ⁻¹ K ⁻¹]	$p/(g\kappa\Pi)$
1000	1004.5000	boundary θ_B	—	—
850	958.9234	interior $\hat{q}$	-48.66	31.63
700	907.1774	interior $\hat{q}$	-67.45	27.53
500	824.0269	interior $\hat{q}$	-67.02	21.65
400	773.1305	interior $\hat{q}$	-55.95	18.46
300	712.1246	interior $\hat{q}$	-48.58	15.03
250	675.9784	interior $\hat{q}$	-38.95	13.19
200	634.2263	interior $\hat{q}$	-77.85	11.25
100	520.2782	boundary θ_T	—	—

$\delta_k = \frac{1}{2}(\Pi_{k+1} - \Pi_{k-1})$ is the half-span of the vertical stencil at level k , negative because Π descends. A ten-level set adds 150 hPa. The unknown vector of the coupled system is therefore (Φ_k, ψ_k) on the seven interior levels, and the sources are the seven $\hat{q}_k$ and the two boundary temperatures.

2 The steps

2.1 Step 1: input and the sphere

Input. Two states on the same northern-hemisphere grid and the same nine levels, an *event* and the *climatology* it is measured against: geopotential height z , temperature T , and the wind u, v , each $(9, n_{\text{lat}}, n_{\text{lon}})$, bottom-up, latitude ascending from the equator, longitude ascending from zero. The walkthrough event is the ERA5 blocking ridge of 8 January 2025, centre 43.5 N 226.5 E, on a 61×240 grid (Figure 1); its 500 hPa height anomaly at the centre is 272 m. Where a level lies below ground, T , u and v are carried down unchanged and the height is continued hydrostatically, $z_k = z_{k+1} + (R_d \bar{T}/g) \ln(p_{k+1}/p_k)$.

The mirror. The operator is global, so the hemisphere is extended to the sphere as an *even* function of latitude, in two passes. The wind is first mirrored as the vector it is, u even and v odd, and its vorticity is taken; that vorticity is then mirrored as an even scalar, and ψ and the rotational wind used downstream are built from it. Height and temperature are mirrored evenly from the outset. With every coefficient and source even, the operator commutes with the reflection $\varphi \rightarrow -\varphi$, the solution is exactly even, and its northern half solves the northern problem under a condition of no meridional gradient at the equator.

The Coriolis parameter. $|f|$ has a corner at the equator that would appear inside ∇f in the balance term; it is smoothed,

$$f^* = 2\Omega\sqrt{\sin^2\varphi + \sin^2\varphi_0}, \quad \varphi_0 = 12^\circ, \quad (13)$$

so that $f^* = 3.0322 \times 10^{-5} \text{ s}^{-1}$ at the equator, the value of $|f|$ at 12 N.

The equatorial weight. The mirrored mean state is continuous but not smooth at the equator. The quadratic terms of the operator are multiplied by the weight

$$w = x^3 (10 - 15x + 6x^2), \quad x = \text{clip}\left(\frac{|\varphi| - 5^\circ}{20^\circ - 5^\circ}, 0, 1\right), \quad (14)$$

which is zero equatorward of 5 N, one poleward of 20 N, and twice differentiable at both ends. It multiplies the *products* the quadratic terms are built from (Step 3), never a source or a solution.

latitude [deg N]	0.5	1	2	5	10	10.5	15	20	30
$f^*/ f $	23.85	11.95	6.04	2.59	1.56	1.52	1.28	1.17	1.08
weight w	0	0	0	0	0.21	0.26	0.79	1.00	1.00

The solver grid. Every product, coefficient and derivative is formed on a Gauss–Legendre grid with no pole row and an even number of latitudes, so no row lies on the equator. For a triangular truncation L the grid has $\lfloor(3L+1)/2\rfloor + 1$ latitudes, rounded up to even, and the next power of two of at least twice that in longitude: the three-halves rule, which represents products of two fields truncated at L without aliasing. Read the other way, a grid of n_{lat} rows carries $L = 2n_{\text{lat}}/3 - 1$: 96×192 gives T63 (the walkthrough), 128×256 gives T84 (the catalogue). Scalars move from the data grid to the solver grid through the spectrum; winds are carried as $u \cos \varphi$ and $v \cos \varphi$, which are band limited where the components are not.

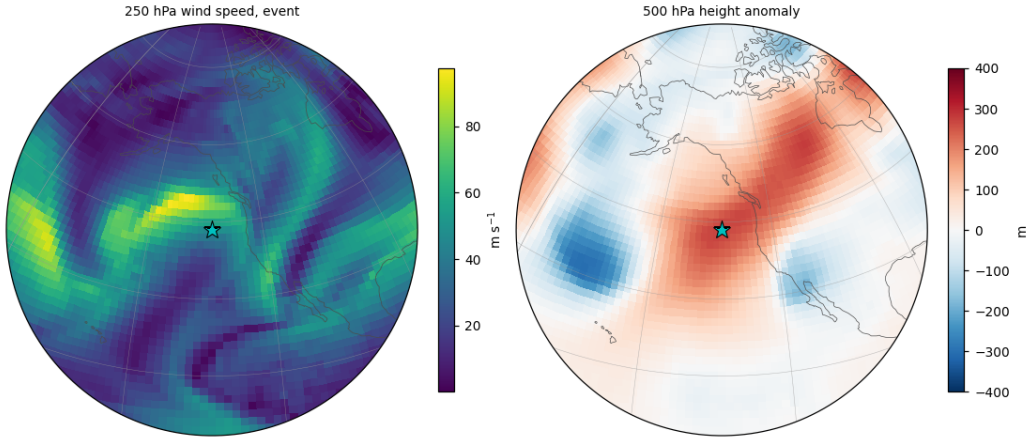


Figure 1: Step 1, the event: 250 hPa wind speed and the 500 hPa height anomaly against the climatology, drawn on a 40-degree disc about the centre. The difference between the two states is what is decomposed.

2.2 Step 2: Pass A/B, the observed state

Streamfunction. The vorticity is formed from the wind by quadrature against the derivatives of the Legendre functions, so the $\tan \varphi$ term of (3) is carried without being evaluated, and the streamfunction is the exact inverse Laplacian, coefficient by coefficient:

$$\psi_{nm} = \frac{\zeta_{nm}}{-n(n+1)/a^2}, \quad n \geq 1, \quad (15)$$

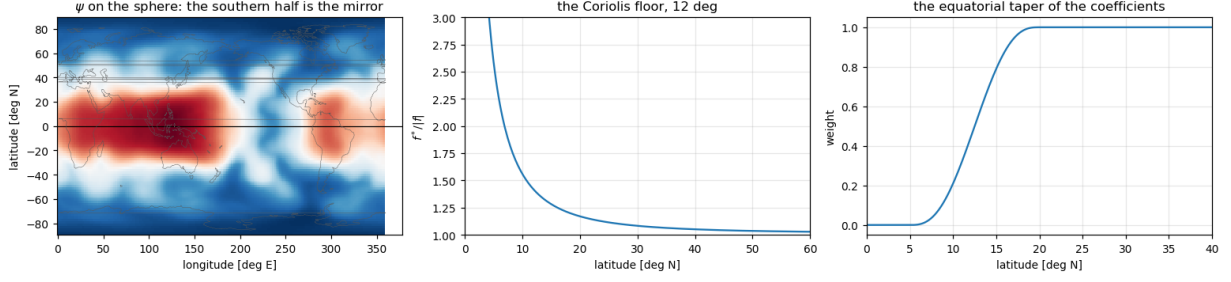


Figure 2: Step 1, the sphere. Left: the 250 hPa streamfunction on the global grid, the southern half the mirror of the northern. Centre: the Coriolis floor as the ratio $f^*/|f|$. Right: the weight w on the operator's quadratic products across the equatorial band.

with the global constant $n = 0$, which carries no wind, set to zero. No relaxation and no boundary value enter. On the walkthrough event the rotational wind at 250 hPa has 31.0 m s^{-1} rms over the hemisphere against 29.8 for the full wind (Figure 3).

Potential vorticity. The source (5) is evaluated from the observed (ψ, Φ) with the operator's own stencils (`pv_source="operator"`, the default), so that the observed pair satisfies the discrete potential-vorticity row exactly:

- Φ_{III} by the three-point non-uniform second difference of the geopotential on all nine levels,

$$\left. \frac{\partial^2 F}{\partial \Pi^2} \right|_k = b_k^{\text{lo}} F_{k-1} + b_k^{\text{c}} F_k + b_k^{\text{hi}} F_{k+1}, \quad b_k^{\text{c}} = \frac{-2}{\delta_k^+ \delta_k^-}, \quad b_k^{\text{hi}} = \frac{2}{\delta_k^+ \delta_k^2}, \quad b_k^{\text{lo}} = \frac{2}{\delta_k^- \delta_k^2}, \quad (16)$$

$\delta_k^+ = \Pi_{k+1} - \Pi_k$, $\delta_k^- = \Pi_k - \Pi_{k-1}$, $\delta_k^2 = \Pi_{k+1} - \Pi_{k-1}$; exact for quadratics in Π , with rows summing to zero;

- the boundary temperatures as the hydrostatic differences of that geopotential across the half levels,

$$\theta_B = -\frac{\Phi_{850} - \Phi_{1000}}{\Pi_{850} - \Pi_{1000}}, \quad \theta_T = -\frac{\Phi_{100} - \Phi_{200}}{\Pi_{100} - \Pi_{200}}; \quad (17)$$

- the shear terms $\partial\psi/\partial\Pi$ and $\partial\Phi/\partial\Pi$ by centred first differences over the interior levels which, at the first and last interior level, reach the operator's ghosts of Step 3 rather than the observed 1000 and 100 hPa fields.

The limited-area convention, plain centred differences of θ and of the wind over the full level set with the boundary temperature as the mean of the two levels beside the half level, is kept as `pv_source="data"`; on the walkthrough event the two differ by 5–14% of the potential vorticity level by level and by 1.4 K rms at the bottom boundary and 2.6 K at the top. The 250 hPa anomaly of the event ranges from -5.59 to 8.02 PVU (Figure 4).

Input checks. A run stops if the geopotential decreases with height at more than 1% of points, if θ decreases with height at more than 5%, or if the geopotential and the temperature are not hydrostatically consistent: the area mean of the boundary temperature the geopotential implies (17), over the mean of the two adjacent levels' θ , must lie in 0.7–1.4 at the top. For the walkthrough event and its climatology alike the ratio is 1.002 at the bottom and 0.995 at the top.

2.3 Step 3: Pass C, the total balanced state

The coupled system (10)–(11), with the observed $\hat{q}$ (floored at 0.01 PVU, the amount added given back to the unclamped points by area weight), θ_B and θ_T as data, is solved for the balanced

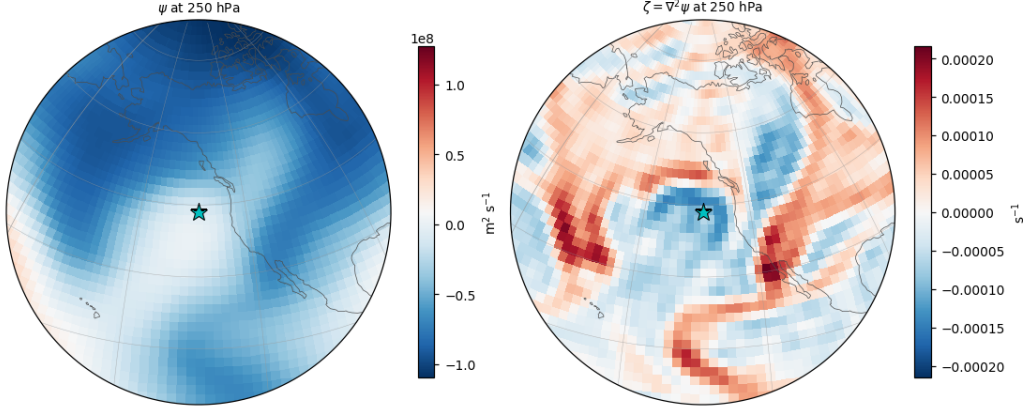


Figure 3: Step 2, the streamfunction recovered from the observed wind at 250 hPa (left) and its Laplacian, the relative vorticity (right).

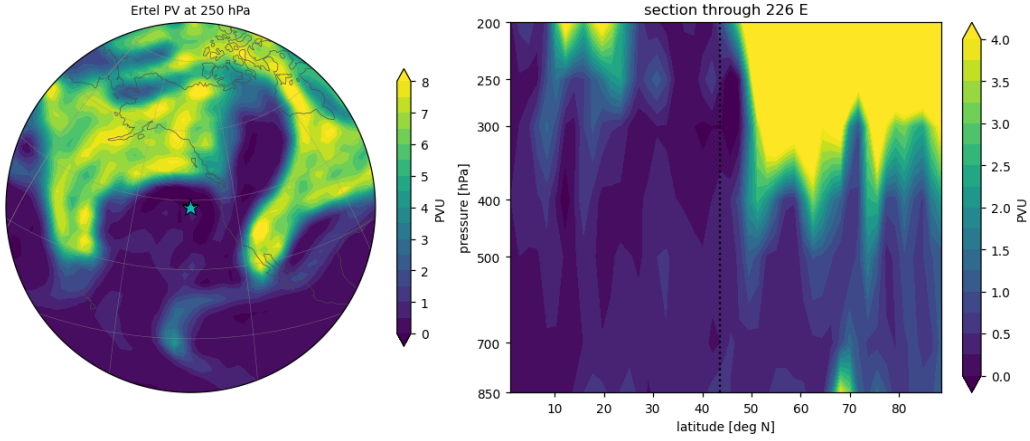


Figure 4: Step 2, Ertel potential vorticity of the event: the 250 hPa map and a vertical section through the centre longitude, the tropopause visible as the crowding of the contours.

$(\Phi_{\text{bal}}, \psi_{\text{bal}})$ on the seven interior levels by Newton’s method. Because both rows are quadratic, the Jacobian is the linearised operator that the pieces of Step 4 use, evaluated at the current iterate.

The linearised operator. At a reference state $(\tilde{\psi}, \tilde{\Phi})$, with $\tilde{\zeta} = \nabla^2 \tilde{\psi}$,

$$M_1[\Phi', \psi'] = \nabla^2 \Phi' - \nabla \cdot (f^* \nabla \psi') - T[\psi'], \quad T[\psi'] = w \tilde{\zeta} \zeta' + w s [B(\tilde{\psi}, \psi') - \tilde{\zeta} \zeta'], \quad (18)$$

$$M_2[\Phi', \psi'] = A \frac{\partial^2 \Phi'}{\partial \Pi^2} + S \nabla^2 \psi' - \nabla \left(\frac{\partial \tilde{\psi}}{\partial \Pi} \right) \cdot \nabla \left(\frac{\partial \Phi'}{\partial \Pi} \right) - \nabla \left(\frac{\partial \tilde{\Phi}}{\partial \Pi} \right) \cdot \nabla \left(\frac{\partial \psi'}{\partial \Pi} \right) + (1 - w) \tilde{\zeta} \langle \Phi'_{\text{III}} \rangle, \quad (19)$$

with $A = f^* + w \tilde{\zeta}$ the absolute vorticity, $S = \langle \tilde{S} \rangle + w(\tilde{S} - \langle \tilde{S} \rangle)$ the static stability $\tilde{S} = \tilde{\Phi}_{\text{III}}$ tapered to its level mean, the reference shear gradients multiplied by w , and s the deformation limiter below. Poleward of 20°N, $w = 1$; with $s = 1$ the rows are (2.10)–(2.11) of Davis and Emanuel (1991) in divergence form: $\nabla^2 \Phi' = \nabla \cdot [(f^* + \tilde{\zeta}) \nabla \psi'] + \nabla \cdot [(\nabla^2 \psi') \nabla \tilde{\psi}] - \nabla^2 (\nabla \tilde{\psi} \cdot \nabla \psi')$. The vertical second difference is (16) with the ghosts folded in: substituting

$$\Phi_0 = \Phi_1 + \theta_B (\Pi_1 - \Pi_0), \quad \Phi_{N-1} = \Phi_{N-2} - \theta_T (\Pi_{N-1} - \Pi_{N-2}) \quad (20)$$

into the end rows folds one coupling weight onto the diagonal and moves the boundary temperature

to the right-hand side as θ_B/δ_1 at the first interior level and $-\theta_T/\delta_{N-2}$ at the last. The centred first differences inside the cross terms reach the same ghosts, the streamfunction's being (12).

The residual. The Newton residual is evaluated through the operator itself,

$$F(x) = M_{x/2}[x] - b_{x/2}, \quad (21)$$

the operator frozen at half the state applied to the whole state, less the right-hand side assembled at the same half state. For a quadratic system this is the exact residual (the midpoint identity of Step 4), and the Jacobian M_x is then the derivative of exactly the system the residual measures, floors, weight and limiter included.

Gauge. A constant added to ψ on any level carries no wind, and a single global constant added to Φ passes through the vertical stencil, so the operator has a null space of dimension $n_{\text{int}} + 1$. The balance row is a sum of divergences, so its $n = 0$ coefficient vanishes identically; that slot is overwritten with the level mean of ψ' , pinned to zero. A rank-one term on the potential-vorticity row, proportional to the column mean of the $n = 0$ coefficient of Φ' with a scale built from f^* and the vertical stencil alone, removes the global constant. The observed state's level-mean geopotential is removed before the first step.

Scaling. The unknown column is $(\Phi, f_0\psi)$ with $f_0 = 10^{-4} \text{ s}^{-1}$ and the rows are $(r_1, (f_0/\bar{S}_k)r_2)$, $\bar{S}_k$ the level-mean static stability, so that a single Krylov tolerance means the same thing on both rows. This is a similarity transform.

Preconditioner. Freezing A and S at their level means A_k, S_k and dropping the cross and deformation terms leaves a system that spherical harmonics diagonalise horizontally, $\lambda_n = -n(n+1)/a^2$, and eliminating ψ exactly gives one tridiagonal problem per harmonic,

$$\left[\text{diag}\left(\frac{A^2}{S}\right) T + \lambda_n I \right] \Phi = \frac{A}{S} r_2 + r_1, \quad \psi = \frac{1}{A} \left(\Phi - \frac{r_1}{\lambda_n} \right), \quad (22)$$

T the folded vertical second difference. The reduced matrix is negative definite for every $n \geq 1$; the inverses are precomputed once per event. The $n = 0$ block uses the rank-one-corrected operator for Φ and reads the streamfunction mean off the balance row.

Linear solve. One preconditioned GMRES per Newton step, restart 60, relative tolerance 10^{-8} , a budget of 800 operator applications; a solve counts as converged only if an independently recomputed $\|b - Mx\|/\|b\|$ is below 10^{-7} .

Floors. $A \geq 10^{-6} \text{ s}^{-1}$ and $S \geq 10^{-4}$ (SI), applied by the **parity** rule: a value below a tenth of the floor is set to the floor. The clamped area fraction per level is reported.

Deformation limiter. Where the reference deformation exceeds the absolute vorticity the linearised balance row is not elliptic. The safety net scales the deformation part of (18) alone,

$$s = \min\left(1, \frac{(1-m)A}{w\tilde{D}}\right), \quad m = 0.05, \quad (23)$$

so the symbol becomes $A \mp s w \tilde{D} \geq mA$ while the vorticity part $\tilde{\zeta}\zeta'$, on which gradient-wind balance rests, is untouched. The limiter is a frozen field, one factor per interior level on the grid. By default (**adaptive**) it starts as one everywhere and is brought in only on evidence of a fold: an inner solve that runs out of its budget, a line search that fails, or stagnation (three or more halvings with a residual not halved). Each refresh takes the pointwise minimum of the limiter in force, the observed state's and the current iterate's; at most six refreshes are allowed and each is counted. It was needed by 2 of the 50 catalogue events, on at most 17% of a level's area. The pieces take the limiter the total inversion converged with.

Line search and stopping. Backtracking by halves, at most eight times, accepting the first step with $\|F(x + \sigma\delta)\| \leq (1 - 10^{-4}\sigma)\|F(x)\|$; if none passes, the best step tried is applied. After

every step the two boundary levels are rebuilt from (20) and (12). The iteration stops when the largest geopotential increment over the interior grid is below $0.981 \text{ m}^2 \text{ s}^{-2}$ ($g \times 0.1 \text{ m}$) and that step's line search succeeded, with a cap of 20 steps. Beside the increment, the report carries the residuals of the equations *as posed* (no weight, floors or limiter) at the returned state: the balance row inverted to a height in metres and the potential-vorticity row in PVU, globally and poleward of 20 N, together with the area fraction per level on which the linearised balance row is not elliptic there.

Numbers. The walkthrough event converges in five steps with increments of 308, 21.6, 1.50, 0.12 and 0.037 m (Figure 5), no backtracking, no inner solve short of tolerance and the limiter never brought in; the posed potential-vorticity equation is met to 0.0040 PVU rms poleward of 20 N, the posed balance equation to $1\text{--}4 \times 10^{-7} \text{ s}^{-1}$ of equivalent vorticity there, and the linearised balance row is not elliptic at the returned state on 1.5% of the area at 850 hPa rising to 14.1% at 300 hPa. Across the fifty catalogue events at T84, 45 converge in four steps, three in five, and the two that meet a fold bring the limiter in once and finish in nine and ten; the potential-vorticity residual poleward of 20 N is 0.0043 PVU rms at the median, 0.0019 to 0.0079 across events.

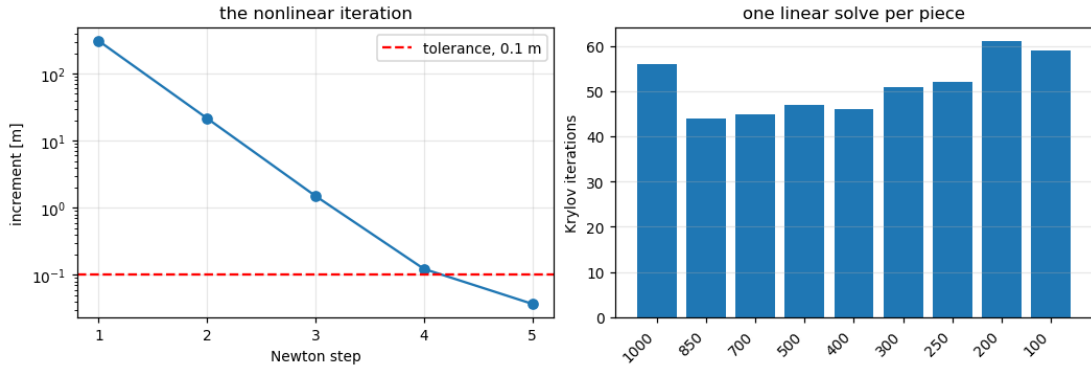


Figure 5: Step 3, the nonlinear iteration on the walkthrough event: the Newton increment in metres of height against the 0.1 m tolerance (left), and the Krylov iterations of the nine piece solves of Step 4 against the one frozen operator (right).

2.4 Step 4: Pass D, the pieces

The frozen operator. The perturbation is measured from the observed climatological mean, $\psi' = \psi_{\text{bal}} - \bar{\psi}$, $\Phi' = \Phi_{\text{bal}} - \bar{\Phi}$, and the operator (18)–(19) is frozen once at the midpoint,

$$\tilde{\psi} = \bar{\psi} + \frac{1}{2}\psi', \quad \tilde{\Phi} = \bar{\Phi} + \frac{1}{2}\Phi', \quad A = f^* + w\tilde{\zeta}, \quad \tilde{S} = \frac{\partial^2 \tilde{\Phi}}{\partial \Pi^2}, \quad (24)$$

its two boundary levels rebuilt as the ghosts (20), (12) of the midpoint temperature $\bar{\theta} + \frac{1}{2}\theta'$. For a quadratic nonlinearity N and a bilinear B the midpoint linearisation is an identity,

$$N(\bar{x} + x') - N(\bar{x}) = DN(\bar{x} + \frac{1}{2}x')[x'], \quad (25)$$

$$B(\bar{x} + x', \bar{y} + y') - B(\bar{x}, \bar{y}) = B(x', \bar{y} + \frac{1}{2}y') + B(\bar{x} + \frac{1}{2}x', y'), \quad (26)$$

so $M_{\bar{x}}$ applied to the full perturbation returns exactly the difference between the nonlinear system at the total state and at the mean. Because w and s multiply symmetric products, the identities hold with the weight and the limiter on. This is the full-linear method of Davis (1992): the pieces of this operator sum to the all-sources solve by construction.

The sources. Nine, one per piece: the potential-vorticity anomaly on each of the seven interior levels, with $\theta_n = 0$ on both boundaries, and the temperature anomaly on each boundary, with

$\hat{q}_n = 0$ in the interior. The interior anomaly is the difference of the event and mean fields floored *separately* at 10^{-5} PVU; the boundary anomalies are plain differences. The right-hand side of a piece is linear in its source:

$$b_1 = 0, \quad b_2 = \hat{q}_n - A \Theta_n - (1 - w) \tilde{\zeta} \langle \Theta_n \rangle + \nabla \left(\frac{\partial \tilde{\psi}}{\partial \Pi} \right) \cdot \nabla \Phi_n^{\text{gh}} + \nabla \left(\frac{\partial \tilde{\Phi}}{\partial \Pi} \right) \cdot \nabla \psi_n^{\text{gh}}, \quad (27)$$

Θ_n the folded boundary source ($\theta_{B,n}/\delta_1, -\theta_{T,n}/\delta_{N-2}$) and $\Phi_n^{\text{gh}}, \psi_n^{\text{gh}}$ the ghost parts of the first differences inside the cross terms, the streamfunction's built by (12) from the piece's own boundary anomaly. The boundary temperature therefore enters twice, through the stability term and through the shear terms, from one vertical operator.

One linear solve per piece. Each piece is one preconditioned GMRES solve against the same operator and the same preconditioner, with the Krylov settings of Step 3; on the walkthrough event the nine pieces take 44 to 61 iterations (Figure 5, right). The solved interior is extended to the two boundary levels hydrostatically: Φ_n by (20) and ψ_n by the thermal-wind ghost (12), both from the piece's own boundary anomaly, zero for an interior piece.

Induced wind. From each piece's streamfunction, spectrally: $u_n = -\partial_y \psi_n$, $v_n = \partial_x \psi_n$, with $u \cos \varphi$ and $v \cos \varphi$ synthesised from a table of Legendre combinations that divides by nothing. For the walkthrough event the rms of the 250 hPa wind induced by each piece over the hemisphere is

piece	1000 θ	850	700	500	400	300	250	200	100 θ
rms [m s ⁻¹]	3.83	2.07	1.80	1.91	3.19	7.21	10.29	11.73	14.66

against an observed anomaly of 21.09 m s⁻¹ (Figure 6). The top-boundary piece is the largest single piece: a planetary-scale temperature anomaly at the lid has a deep, nearly barotropic response, part of which cancels against the interior upper-level pieces.

2.5 Step 5: grouping and the scale split

Groups. Pieces are linear in their sources, so groups are sums of pieces, exactly, formed after the solve:

group	sources
surface	1000 hPa boundary potential temperature
lower	potential vorticity at 850, 700, 500 hPa
upper	potential vorticity at 400, 300, 250, 200 hPa and the 100 hPa boundary

On the walkthrough event they come to 3.83, 3.68 and 20.68 m s⁻¹ rms at 250 hPa (Figure 7).

Planetary and eddy. With `-pieces scale` the upper source is split further by horizontal scale into `upper_p` and `upper_e`, and the split is applied to the *source*, additively, so that the parts still sum to the piece they came from:

1. the upper-level anomaly (400–200 hPa) is filtered on the whole circle to zonal orders 1–4, a selection of spherical-harmonic orders;
2. the object is seeded at the most negative filtered point within $\pm 10^\circ$ latitude and $\pm 20^\circ$ longitude of the tracked centre, and the contour is 0.35 of the local minimum there;
3. the object is the three-dimensional six-connected component of the filtered field below that contour holding the seed, sought inside the event's $\pm 30^\circ \times \pm 60^\circ$ box, whose columns run around the circle;

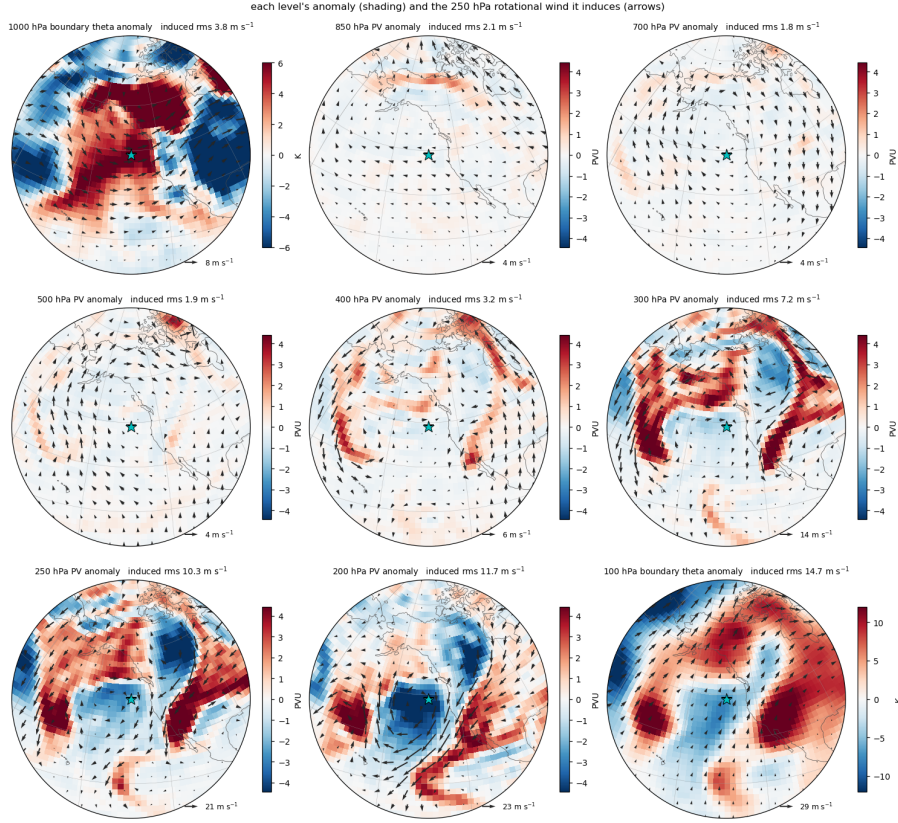


Figure 6: Step 4, the wind at 250 hPa induced by each source. Shading is the source that drives the piece, potential vorticity on the seven interior levels and potential temperature on the two boundaries; arrows are the rotational wind it induces at 250 hPa, with the arrow scale set panel by panel.

4. the masked field is filtered again to orders 1–4; that is the planetary part, and the eddy part is the remainder of the upper anomaly;
5. the top boundary temperature is split the same way, with the mask of the highest interior level.

The four pieces `surface`, `lower`, `upper_p`, `upper_e` then exhaust the sources; the contour, the local minimum and the object's area fraction are written to the output metadata.

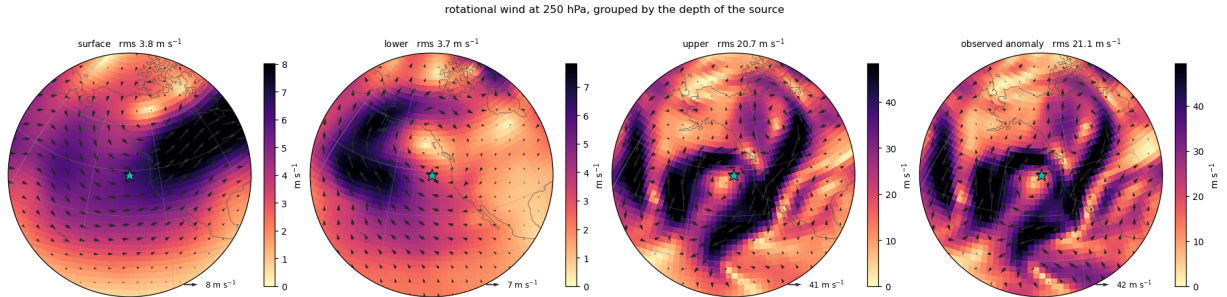


Figure 7: Step 5, the pieces grouped into surface, lower and upper contributions to the 250 hPa rotational wind, beside the observed anomalous rotational wind. Colour and arrow scales are set panel by panel; the rms in each title is what to compare.

2.6 Step 6: closure and residual

Closure. The sum of the nine pieces is compared with the solve that has all nine sources present at once; the two agree to rounding, because every piece is $M^{-1}b_n$ against one operator and $\sum_n M^{-1}b_n = M^{-1}\sum_n b_n$. On the walkthrough event the relative difference is 1.7×10^{-8} , on the 79 N event of Step 7 5.1×10^{-9} . This is checked on every run.

Residual. The sum of the pieces against the *observed* anomalous rotational wind leaves the part the balance equations do not represent: the anomaly of the unbalanced and divergent flow. It is not a wall response, there being no wall, and it is stored under its own name. On the walkthrough event the pieces sum to 20.01 m s^{-1} rms at 250 hPa over the hemisphere against an observed anomaly of 21.09, leaving 3.10 m s^{-1} , 15% of the anomaly (Figure 8); at 79 N the figures are 19.10 against 20.25, leaving 4.17, or 21%. Over the fifty catalogue events, on the common patch at 400–200 hPa, the residual is 0.26 m s^{-1} of a 10.12 m s^{-1} anomaly, 2.6%.

What the pieces close against. Three quantities are kept apart. The pieces sum to the all-sources linear solve exactly. They differ from the balanced perturbation $\psi_{\text{bal}} - \bar{\psi}$ by the response to the mean state’s own imbalance, because the mean enters as observed rather than balanced: 4.14 m s^{-1} rms at 250 hPa on the walkthrough event. And the balanced total state differs from the observed rotational wind by the event’s own departure from balance: 4.97 m s^{-1} out of 30.99. The residual above is the difference of the last two.

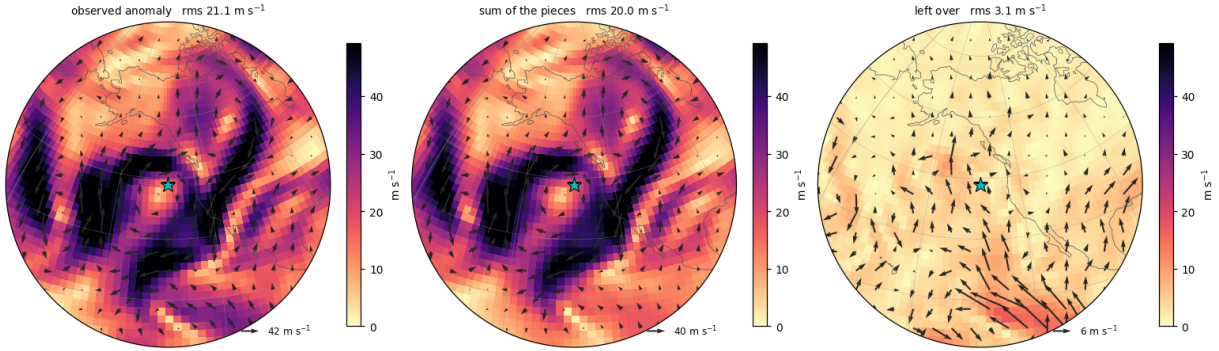


Figure 8: Step 6, the observed anomalous rotational wind at 250 hPa (left), the sum of the nine pieces (centre), and what is left over, the unbalanced and divergent part (right).

2.7 Step 7: output on any patch

Nothing in Steps 1–6 refers to where the event is: the operators are global, the grid has no pole row and no metric factor is divided by. The Greenland event at 79 N that ships with the package is inverted by the same calculation in five Newton steps to a final increment of 0.003 m (Figure 9). Two croppings are written.

The geographic patch. An event-centred box of half-widths $30^\circ \times 60^\circ$ on the data’s own regular grid, longitudes wrapped. Rows past the pole are NaN, because a column of the box carries one longitude and the far side of the pole belongs to another. The pole row itself is NaN ($|\cos \varphi| \leq 10^{-8}$): the wind is defined there but its eastward and northward components are not. Rows south of the data’s first row, the mirror scaffold, are NaN as well. The two-dimensional fields are the column average over 400–200 hPa weighted by $\exp(-z/H)$, $H = 7000 \text{ m}$, divided by the weight of the valid levels only.

The rotated patch (-rotated-track). The frame is rotated so that the centre sits on its equator; the wind is carried as its Cartesian components, which are smooth across the pole, sampled by cubic interpolation on the rotated grid and rotated into the local frame. The patch is complete

for any centre, the pole included, and has the same shape for every event. Figure 11 shows both croppings of two events at 55 and 80 N.

The key contract. One compressed `npz` per event, keys named so that a loader enumerating pieces by the prefix `u_rot_anom_ppvi_` counts exactly the pieces:

key	content
<code>u/v_rot_anom_ppvi_{piece}, ..._3d</code>	wind induced by one piece: the 400–200 hPa column average, and all nine levels; <code>piece</code> is 1000, 850, ..., 100 or <code>surface</code> , <code>lower</code> , <code>upper_p</code> , <code>upper_e</code>
<code>u/v_rot_anom_residual_ppvi, ..._3d</code>	observed anomaly less the sum of the pieces: the unbalanced part
<code>u/v_rot_anom, ..._3d</code>	the observed anomalous rotational wind the pieces account for
<code>pv_anom_wu_3d</code>	the potential-vorticity anomaly in SI, boundary levels NaN
<code>lat_vec, lon_vec, lat_rel, lon_rel</code>	patch axes, absolute and relative to the centre
<code>..._3d_rot, lat_rel_rot, lon_rel_rot</code>	the rotated-frame patch, when requested
<code>meta_*</code>	<code>pieces_mode</code> , <code>pv_source</code> , <code>deformation_limiter</code> , <code>centre_lat/lon</code> , <code>levels</code> , <code>level_pressures</code> , <code>piece_names</code> , <code>solver_grid</code> , <code>lmax</code> ; <code>newton_steps</code> , <code>newton_converged</code> , <code>newton_final_increment_m</code> , <code>newton_final_balance_m(_extratropics)</code> , <code>newton_final_pv_pvu(_extratropics)</code> , <code>newton_final_pv_pvu_rms_extratropics</code> , <code>newton_deformation_fraction</code> , <code>newton_limiter_refreshes</code> , <code>newton_final_nonelliptic_fraction</code> , <code>piece_deformation_fraction</code> , <code>linear_iterations</code> , <code>inner_solves_unconverged</code> , <code>all_pieces_converged</code> ; <code>clamp_worst_fraction</code> , <code>pv_floor_fraction_event/mean</code> ; with <code>-pieces</code> scale also <code>split_q_min</code> , <code>split_contour</code> , <code>split_object_fraction</code> , <code>split_top_fraction</code>

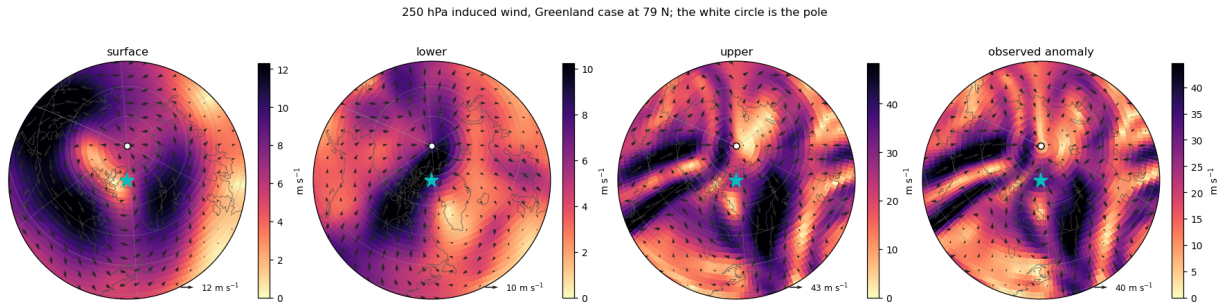


Figure 9: Step 7, the same calculation for the Greenland event at 79 N: the 250 hPa wind induced by the surface, lower and upper groups, and the observed anomaly, on a 40-degree disc about the centre. The white circle is the North Pole, an ordinary point inside every panel.

2.8 Step 8: a catalogue

```
pvinv-sph catalogue.csv --out-dir results/ --workers 25 --pieces scale
```

The catalogue is a CSV with columns `event_id`, `state_path`, `state_index`, `clim_path`, `clim_index`, `lat`, `lon`, checked before the run starts. Each event is one process with the linear algebra held to one BLAS thread; the file is written to a temporary name and renamed, so an

interrupted run leaves no half file, and `-skip-existing` resumes it. A failure is recorded per event rather than raised. The fifty winter blocking peaks of the composite example, at T84 on a 128×256 grid, took 12.0 minutes on 25 workers, 50 of 50 converged, 4 Newton steps at the median; one event takes 90–130 s at T84 on one core, 3.2 minutes in the truncation sweep of that example.

3 Old and new side by side

The windowed chain (the Wu/Davis Fortran driven through `pvtend.ppvi`) and this package solve the same system (10)–(11) with the same boundary conditions and the same midpoint linearisation. The Fortran alternates a two-dimensional streamfunction relaxation per level with a three-dimensional geopotential relaxation; here the coupled system is solved at once by Newton–Krylov. Same solution, different solver, domain and discretisation:

	windowed chain	this package
domain and lateral boundary	fixed band 10.5–85.5 N, a $\pm 90^\circ$ longitude window about the event; homogeneous Dirichlet walls; lateral condition <code>ibc</code> 0 (homogeneous), 1 (full perturbation on the walls) or 2 (nested per-piece walls); the wall response carried as a piece of its own	the closed sphere: no wall, no wall piece, no <code>ibc</code> ; the hemisphere mirrored evenly with f^* , so the northern half solves the northern problem under a Neumann equator
horizontal discretisation	five-point latitude–longitude finite differences with $\cos \varphi$ factors	spherical harmonics on a Gaussian grid (T63, T84); every metric term inside the transforms, no $\tan \varphi$ or $1/\cos \varphi$ evaluated
balance equation	plane form $\nabla \cdot (f \nabla \psi) + 2(\psi_{xx}\psi_{yy} - \psi_{xy}^2)$	exact spherical form (7), which carries the curvature term $- \nabla \psi ^2/a^2$ of (8), about 2% of the balance terms in a mid-latitude jet
potential-vorticity source	centred differences of θ and of the total wind shear over the full level set; boundary θ the mean of the two levels beside the half level	the operator’s own stencils applied to z : (16) for Φ_{III} , shear terms with the ghosts, boundary θ by (17); the data satisfy the potential-vorticity row exactly (5–14% apart level by level on the walkthrough event)
Pass C solver	alternation: a two-dimensional ψ Poisson sweep per level with the geopotential held, then a three-dimensional Φ SOR sweep, relaxation factors 1.4, the coupling lagged with <code>PART</code> = 0.05 under-relaxation between sweeps; a pass is converged when no point changes by more than a threshold in a sweep, a test on the change rather than on the residual	Newton–Krylov on the coupled system: the Jacobian is the piecewise operator, one preconditioned GMRES per step, line search; converged on the increment (< 0.1 m after a successful line search) with the residuals of the posed equations reported in metres and PVU
Pass D	SOR per piece with the discrete elimination weights <code>ASI</code> , <code>BSI</code> , <code>APHI</code> , <code>BI</code> , which carry the five-point diagonal; <code>INLIN</code> = 1 midpoint coefficients	one GMRES per piece against the same frozen operator; the midpoint linearisation exact for the quadratic system, so the pieces sum to the all-sources solve to 10^{-8}

	windowed chain	this package
boundary ghosts	the same θ' ghost for H and S , with a fixed f_0	$\Phi: \partial\Phi/\partial\Pi = -\theta; \psi: \partial\psi/\partial\Pi = -(\theta - \langle\theta\rangle)/f^*$ with the local f^*
ellipticity devices	$AVO < 0.001 \rightarrow 0.01$, $STB < 0.001 \rightarrow 0.01$, $BI > 0 \rightarrow 0$, in the scaled units	floors in SI, $A \geq 10^{-6} \text{ s}^{-1}$, $S \geq 10^{-4}$; $\hat{q} \geq 0.01$ PVU for the total inversion with area-weighted give-back; the deformation limiter (23) on the balance row's deformation part, on demand
mean state	observed	observed
poles	the band ends at 85.5 N; the five-point stencil changes sign past 89 N and a rectangle continued past the pole folds	ordinary points; the geographic patch blank past the pole and on the pole row, the rotated patch complete
precision	REAL*4	float64
cost	SOR sweeps over levels and pieces, up to 10 000 per pass, one core per event	4 Newton steps at the median of fifty events; 90–130 s per event at T84 on one core, 3.2 min in the truncation sweep; fifty events in 12 min on 25 workers

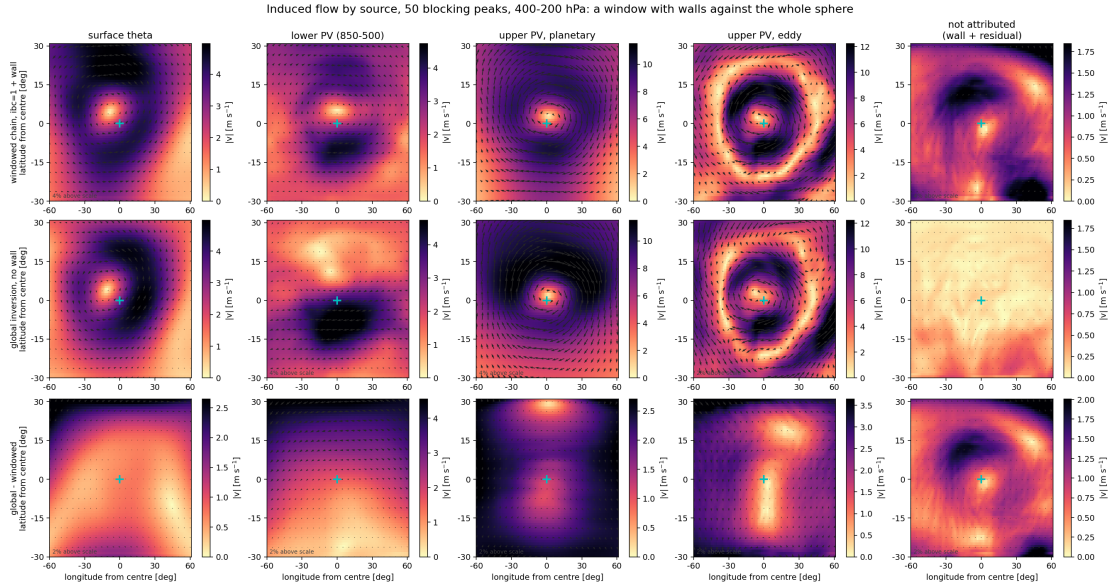


Figure 10: Fifty blocking peaks, composite of the induced flow at 400–200 hPa on the event-centred patch. Columns: the surface, lower, upper-planetary and upper-eddy pieces, and the part not attributed to any source. Rows: the windowed chain (`ibc=1`, wall piece included in the last column), the global inversion, and their difference.

Fifty events. Both chains decompose the same balanced perturbation into the same four sources (Figure 10). On the common patch, rms in m s^{-1} , the windowed chain attributes 3.45 to the surface, 2.66 to the lower levels, 7.56 to the planetary and 6.74 to the eddy upper piece, and leaves 1.09 unattributed, of which 0.79 is its wall piece and 0.99 its residual, against a reference anomaly of 10.14. The global inversion attributes 3.03, 2.61, 8.65 and 6.86 and leaves 0.26, 2.6% of its 10.12 anomaly. Removing the boundary moves the flow from the unattributed part and the surface into the planetary upper-level piece; the eddy and lower pieces barely change.

Two events near the pole. For the control at 55 N and the event at 80 N (Figure 11) the windowed chain cannot reach 24% and 49% of the disc, and misses the observed anomaly by 5.0

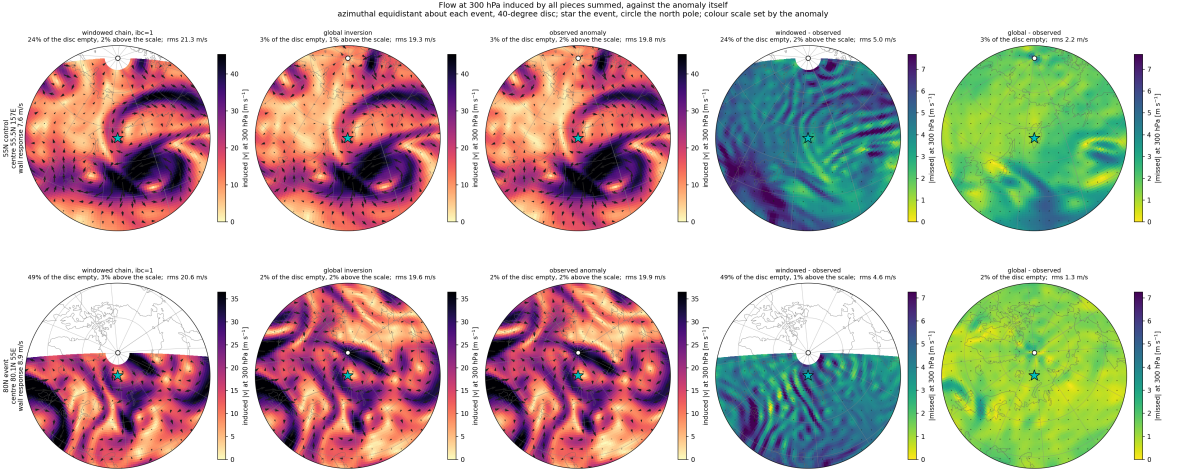


Figure 11: Two events, at 55 N and 80 N: the 300 hPa flow induced by all pieces summed, for the windowed chain and the global inversion, beside the observed anomaly and the two differences from it, on a 40-degree disc about each centre. Star: the event; white circle: the North Pole.

and 4.6 m s^{-1} rms where it does reach. The global inversion converges in four Newton steps for both, leaves 3% and 2% of the disc empty, the pole row, and misses by 2.2 and 1.3 m s^{-1} .

4 Running it

Install. `pip install -e .`; NumPy, SciPy (1.12 or newer, for the `rtol` keyword of the Krylov drivers), `threadpoolctl`, `xarray` and `netCDF4`; `matplotlib` and `cartopy` for the figures only.

Input contract. Each state is four arrays of shape $(n_{\text{lev}}, n_{\text{lat}}, n_{\text{lon}})$ over the northern hemisphere: `lat` ascending from the equator to 90 N, either on the equator or half a step from it; `lon` ascending from 0; `level` descending in pressure, 1000 hPa first; `z` a geopotential *height* in metres; `t` a *temperature* in kelvin; `u`, `v` in m s^{-1} . The climatology is on the same grid and levels. The geopotential and the temperature must be hydrostatically consistent (Step 2).

One event, from Python.

```
from pvinv_sph.config import InversionConfig, MirrorConfig
from pvinv_sph.levels import build_levels
from pvinv_sph.passd import invert_pieces
from pvinv_sph.prepare import prepare_state
from pvinv_sph.sht import SHT, gaussian_grid
from pvinv_sph.sphere import SphereOps
from pvinv_sph.winds import rotational_wind_stack

levels = build_levels("NL9")
solver = SphereOps(SHT(gaussian_grid(96, 192), lmax=63))
cfg = InversionConfig(mirror=MirrorConfig(blend=True))

event = prepare_state(z_ev, t_ev, u_ev, v_ev, lat, lon, levels, solver)
clim = prepare_state(z_cl, t_cl, u_cl, v_cl, lat, lon, levels, solver)
result = invert_pieces(solver, levels, clim, event, cfg=cfg)

# the rotational wind induced by the 300 hPa potential-vorticity anomaly
u, v = rotational_wind_stack(solver, result.pieces["300"].psi_spec)
```

`result.pieces` is keyed by pressure in hPa, "1000" and "100" being the boundary pieces;

`result.diagnostics` carries the Newton history, the Krylov count per piece, the posed-equation residuals and the limiter and clamp reports; `result.summed_psi()` is compared with `all_sources_inversion` for the closure of Step 6. For the patch cropping and the output arrays use `invert_event` from `pvinv_sph.pipeline`.

A catalogue, from the command line. The command of Step 8, with the flags

flag	default	meaning
-pieces	per_level	per_level or scale (Step 5); disjoint key sets, not to be mixed in one directory
-pv-source	operator	stencils of the potential-vorticity source (Step 2)
-deformation-limiter	adaptive	adaptive, observed or off (Step 3)
-krylov-maxiter	800	operator applications per linear solve
-newton-max-steps	20	cap on the nonlinear iteration
-blend-south, -blend-north	5.0, 20.0	edges of the equatorial weight, degrees
-f-floor-deg	12.0	latitude scale of the Coriolis floor
-rotated-track	—	also write the rotated-frame patch
-solver-nlat, -solver-nlon	128, 256	the Gaussian solver grid
-lat-half, -lon-half	30.0, 60.0	output patch half-widths
-workers, -skip-existing	CPU count, —	processes; resume a run

A run that hits the Newton cap is reported, not raised: read `meta_newton_final_increment_m` and the posed-equation residuals before treating it as a failure. A solution quoted equatorward of about 20 N lies inside the Coriolis floor and the weight, and is to be checked against a run with a smaller `-f-floor-deg` and a narrower band.

Tests and figures.

```
PYTHONPATH=src python -m pytest tests -q
python examples/01_walkthrough/build.py
jupyter nbconvert --to notebook --execute --inplace \
    --ExecutePreprocessor.timeout=3600 \
    examples/01_walkthrough/01_walkthrough.ipynb
```

The tests check the vertical stencil, the boundary folding, the balance term against solid-body rotation and under rotation of a vortex to the pole, the spectral round trip, the midpoint identity with the weight on, the limiter and its Jacobian, the pieces' thermal-wind boundary levels, and the closure of the pieces against the all-sources solve. The notebook produces every figure of Section 2.

References

- [1] Charney, J., 1955: The Use of the Primitive Equations of Motion in Numerical Prediction. *Tellus*, **VII**(1), 22–26.
- [2] Davis, C. A., and K. A. Emanuel, 1991: Potential Vorticity Diagnostics of Cyclogenesis. *Monthly Weather Review*, **119**(8), 1929–1953.
- [3] Davis, C. A., 1992: Piecewise Potential Vorticity Inversion. *Journal of the Atmospheric Sciences*, **49**(16), 1397–1411.